Music Genre Classification

CHARITHA BV – IIS20012



References

1. http://marsyas.info/downloads/datasets.html (Dataset - Audio Files)
2. http://cs229.stanford.edu/proj2019spr/report/3.pdf (Feature Extraction)
3. https://www.irjet.net/archives/V6/i5/IRJET-V6I5174.pdf (Feature Extraction)
4. https://medium.com/@sdoshi579/classification-of-music-into-different-genres-using-keras-82ab5339efe0 (Code Reference)

Problem Formulation

Objective: Classification of Music genres (10 different genres)

Dataset details:

http://marsyas.info/downloads/datasets.html

The dataset considered here is GTZAN genre collection. It consists of 10 different genres and each consisting of 100 audio clips which plays for 30 seconds.

Assumptions:

From each music genre, 10 samples are considered

Feature Extraction

- The librosa API is used to extract wave files(.wav)
- Here, two librosa methods are used to extract raw data from the wave files.
  - MFCC (Mel-Frequency Cepstral Coefficients)
  - Chromagram

PCA –Comparison of Sum of the upper triangular matrices

- The value of upper triangular matrices before applying PCA is -173.79940991701517
- The value of upper triangular matrices after applying PCA is 4.937684067143513e-12
- These values shows that after applying PCA the features are independent

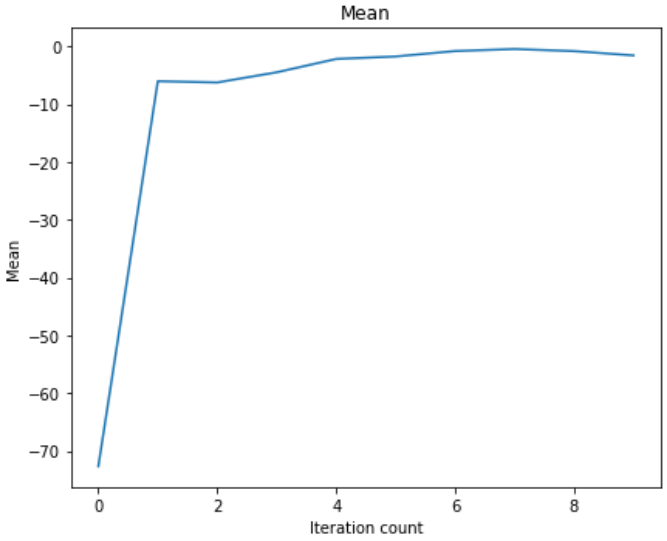
(Theory) GMM EM

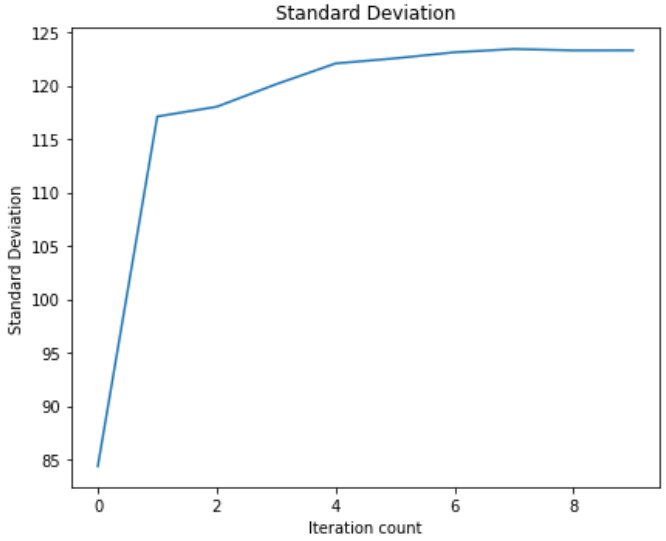
- Gaussian Mixture Models (GMMs) assume that there are a certain number of Gaussian distributions , and each of these distributions represent a cluster .
- Gaussian Mixture Model tends to group the data points belonging to a single distribution together.
- Gaussian Mixture Models are probabilistic models and use the soft clustering approach for distributing the points in different clusters

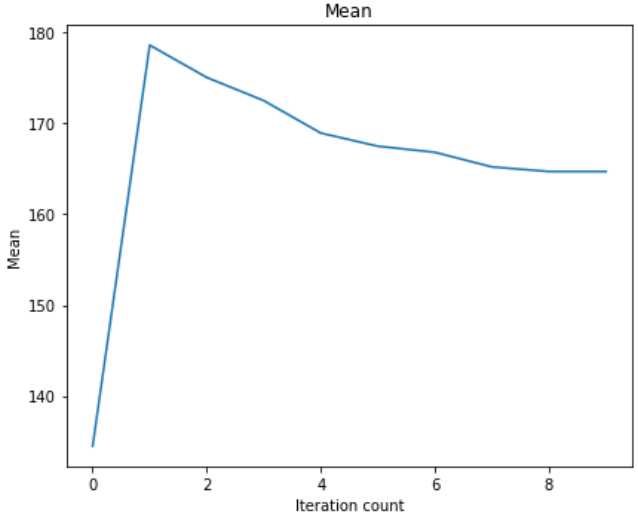
(Theory) GMM EM

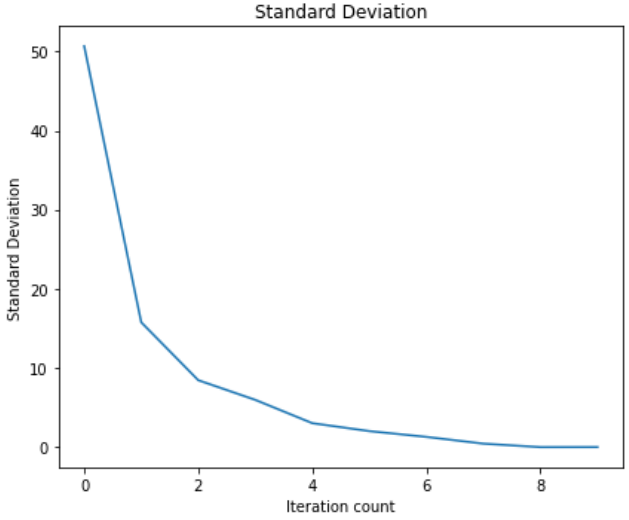
- EM algorithm models the data as being generated by mixture of Gaussians.
- The EM algorithm estimates the parameters of (mean and covariance matrix) of each Gaussian.
- Each Gaussian defines a single cluster of data.
- EM is an iterative algorithm which is used to find the maximum likelihood when there are hidden variables

GMM EM application (100 samples)









Inference

- *How many features (columns) you played with?*

  Two features

- *How to select initial values?*

  Initial values are selected at random.

- *How many iterations?*

  10 iterations were executed.

- *How to choose number of Gaussians?*

  Using Maximum Likelihood Estimate

(Theory) Logistic regression

- Logistic Regression is used when the dependent variable(target) is categorical.
- For example,

  To predict whether an email is spam (1) or (0)

Reference

- https://towardsdatascience.com/logistic-regression-detailed-overview-46c4da4303bc - Logistic Regression

(Theory) Logistic regression

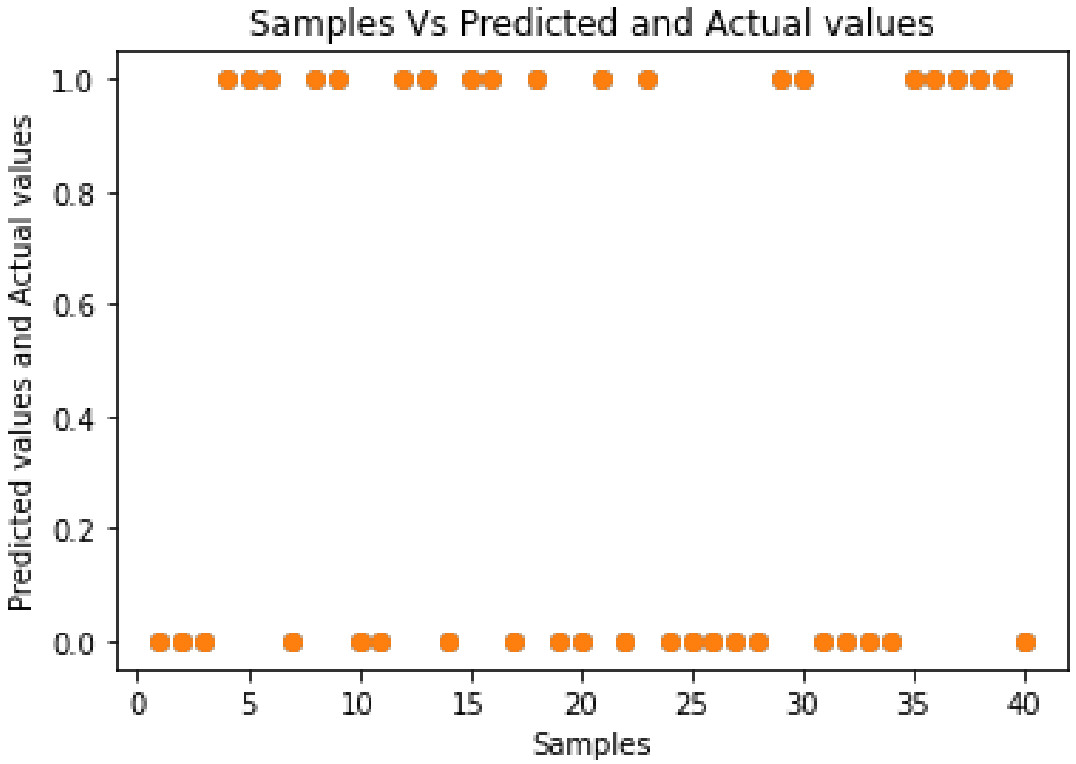
- Training 80% of data - **160 samples**
- Testing remaining 20% of data – **40 samples**

Classification – Two classes are considered

- Class 1 – 'blues'
- Class 2 – 'classical'

Inference

- The prediction Accuracy is 100%

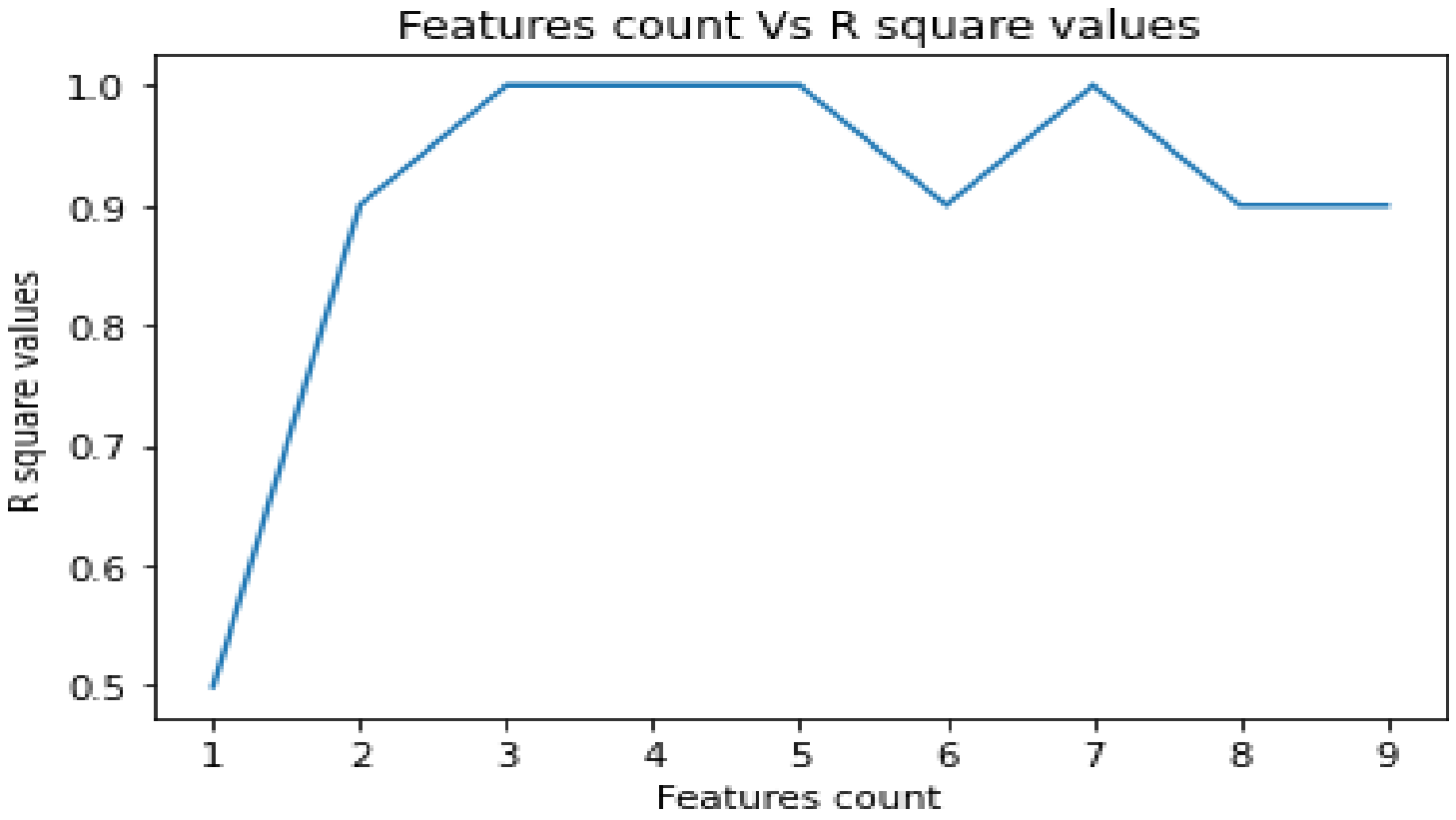


Inference

- *How many features (columns) you played with? What is R^2 values in each case? Plot the graph Y axis is R^2 values*

- ***R square value = 1.0 for the features 3,4,5 and 7. And this is indicates the perfect fit***
- ***It is observed that if number of features increases R square value decreases.***

Inference

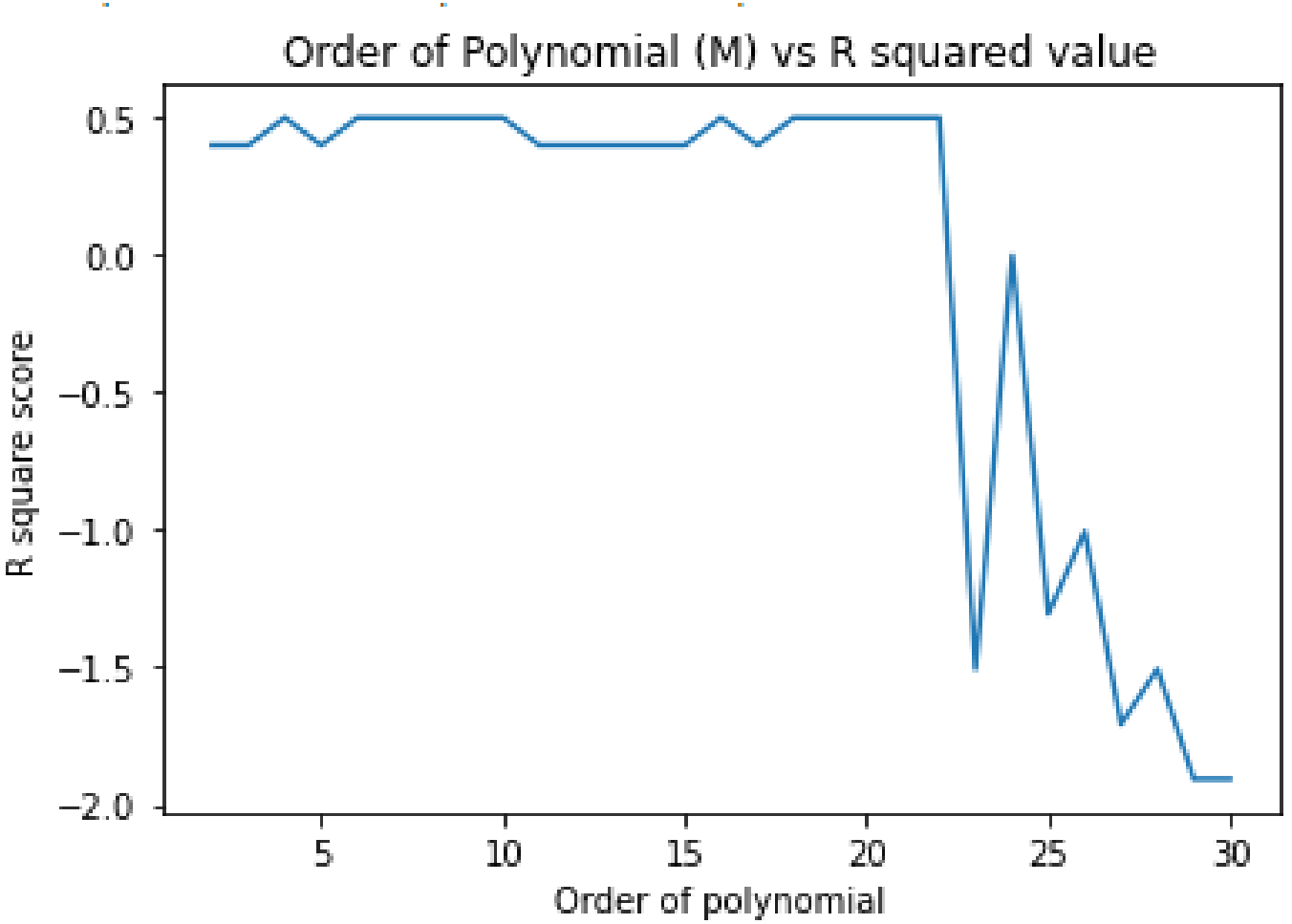

Features count Vs R square values
R square values
1.0
0.9
0.8
0.7
0.6
0.5
1
2
3
4
5
6
7
8
9
Features count

Inference

- *What is the order of polynomial you have selected? ?What is R^2 values in each case? Plot the graph.Y axis is R^2 values*

- ***As the order of polynomial increases, R square values decreases***
- ***The maximum R square value is 0.4987 when the order of polynomials are 2, 4, 5, 6, 7, 8.***

Inference


Order of Polynomial (M) vs R squared value
R square score
0.5
0.0
−0.5
−1.0
−1.5
−2.0
5
10
15
20
25
30
Order of polynomial

References

K-means Clustering

- https://www.analyticsvidhya.com/blog/2019/08/comprehensive-guide-k-means-clustering/

(Theory) k-means Clustering

- K-means clustering is an algorithm which tries to minimize the distance of the points in a cluster with their centroid.
- The main objective of the K-Means algorithm is to minimize the sum of distances between the points and their respective cluster centroid.

(Theory) K-means Algorithm

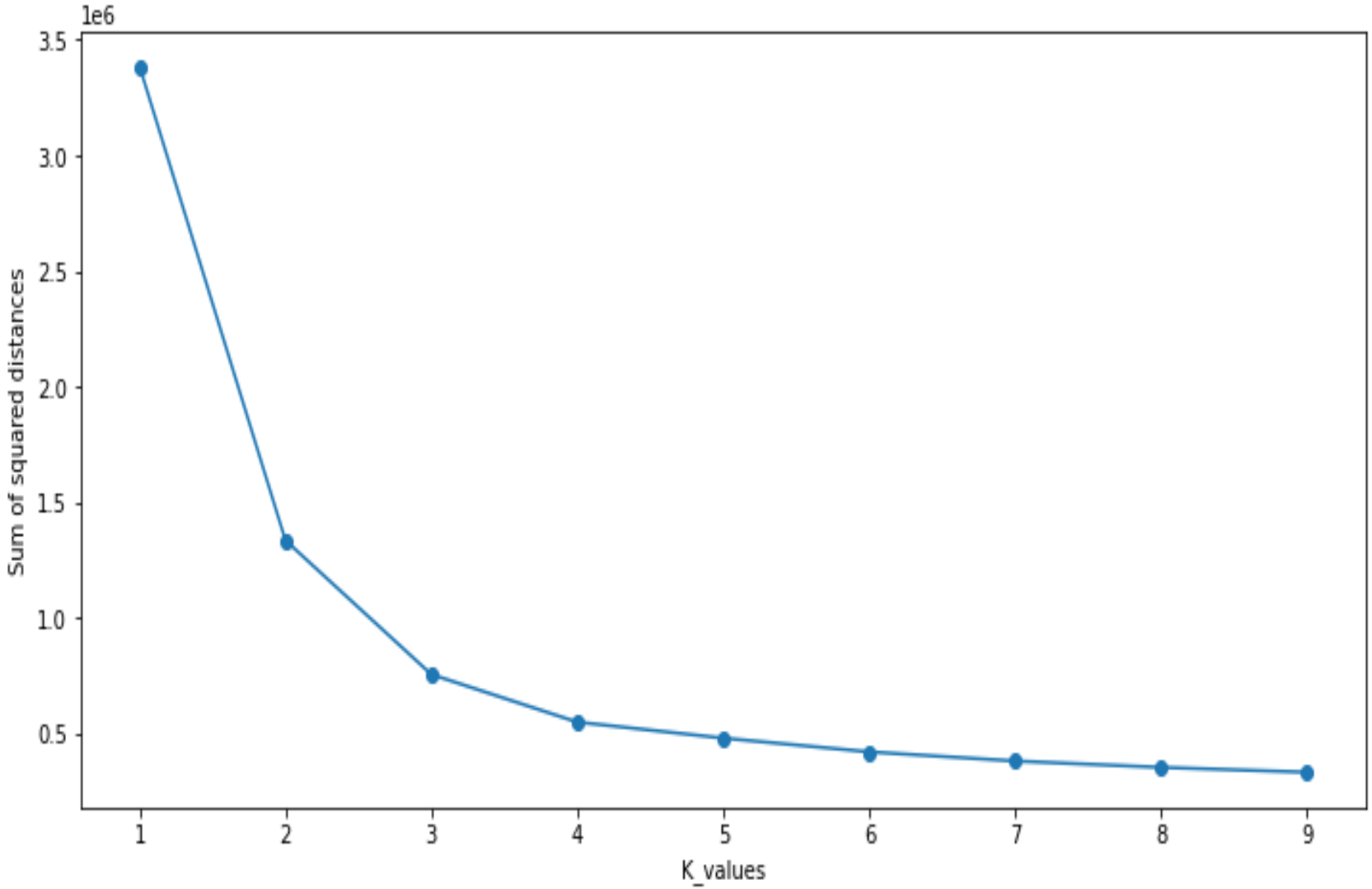
- Step 1: Choose the number of clusters k
- Step 2: Select k random points from the data as centroids
- Step 3: Compute the distances for each data point and cluster centroid
- Step 4 : Assign all the points to the closest cluster centroid
- Step 5: Recompute the centroids of newly formed clusters
- Step 6: Repeat the steps 3 to 5 until the convergence is achieved.

Convergence criteria

The termination conditions are

- Number of maximum iteration permissible.
- No change of centroid values in any cluster.
- No significant movements of sample from one cluster to another.

Inference


1e6
3.5
3.0
2.5
2.0
1.5
1.0
0.5
Sum of squared distances
1
2
3
4
5
6
7
8
9
K_values

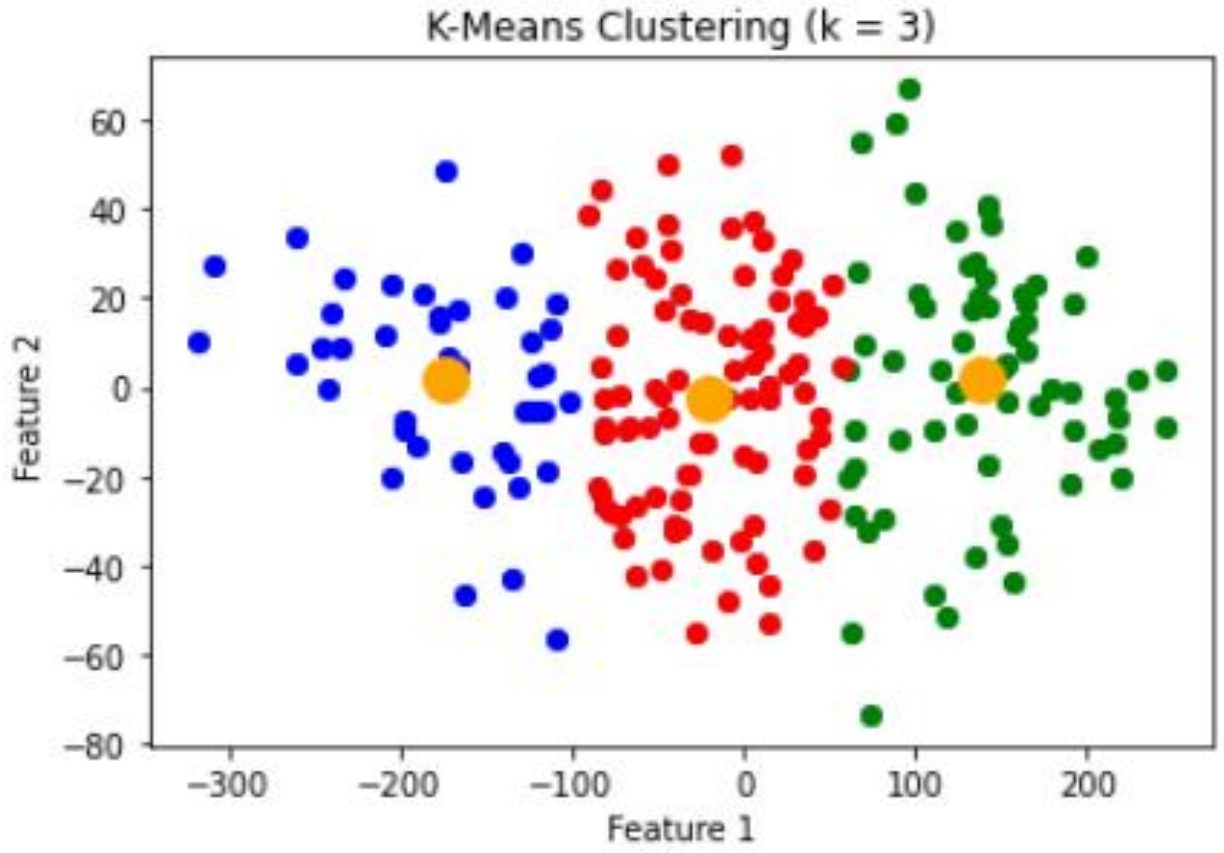
Inference

Use features of your dataset for clustering into *n* clusters. Where n must be learnt by elbow method.

- **The optimal number of clusters is given by the sharp turn in the Elbow graph.**
- **Here, the optimal cluster is 3.**

Inference

- The yellow dots represent cluster centers



(Theory) Neural Network

- Neural networks, also known as artificial or simulated neural networks (ANNs/SNNs) are a subset of machine learning and are at the heart of deep learning algorithms.
- Their name and structure are inspired by the human brain, mimicking the way that biological neurons signal to one another.
- Artificial neural networks (ANNs) are comprised of a node layers, containing an input layer, one or more hidden layers, and an output layer.
- Each node or artificial neuron connects to another and has an associated weight and threshold.

(Theory) Neural Network

- If the output of any individual node is above the specified threshold value, that node is activated, sending data to the next layer of the network.
- Otherwise, no data is passed along to the next layer of the network.
- Neural networks rely on training data to learn and improve their accuracy over time.

Types of neural networks

- Feedforward neural networks or multi-layer perceptrons (MLPs)
- Recurrent neural networks (RNNs)

Inference

- Sigmoid layer network

The number of epochs given is 700 and the accuracy achieved is 65%

```
Epoch 693/700
80/80 [==============================] - 0s 2ms/step - loss: 0.0029 - categorical_accuracy: 0.9719 - mse: 0.0029
Epoch 694/700
80/80 [==============================] - 0s 2ms/step - loss: 0.0025 - categorical_accuracy: 0.9759 - mse: 0.0025
Epoch 695/700
80/80 [==============================] - 0s 2ms/step - loss: 0.0027 - categorical_accuracy: 0.9762 - mse: 0.0027
Epoch 696/700
80/80 [==============================] - 0s 1ms/step - loss: 0.0027 - categorical_accuracy: 0.9739 - mse: 0.0027
Epoch 697/700
80/80 [==============================] - 0s 1ms/step - loss: 0.0017 - categorical_accuracy: 0.9841 - mse: 0.0017
Epoch 698/700
80/80 [==============================] - 0s 1ms/step - loss: 0.0034 - categorical_accuracy: 0.9686 - mse: 0.0034
Epoch 699/700
80/80 [==============================] - 0s 1ms/step - loss: 0.0026 - categorical_accuracy: 0.9744 - mse: 0.0026
Epoch 700/700
80/80 [==============================] - 0s 1ms/step - loss: 0.0027 - categorical_accuracy: 0.9741 - mse: 0.0027
7/7 [==============================] - 0s 2ms/step - loss: 0.0601 - categorical_accuracy: 0.6500 - mse: 0.0601

categorical_accuracy: 65.00%
```

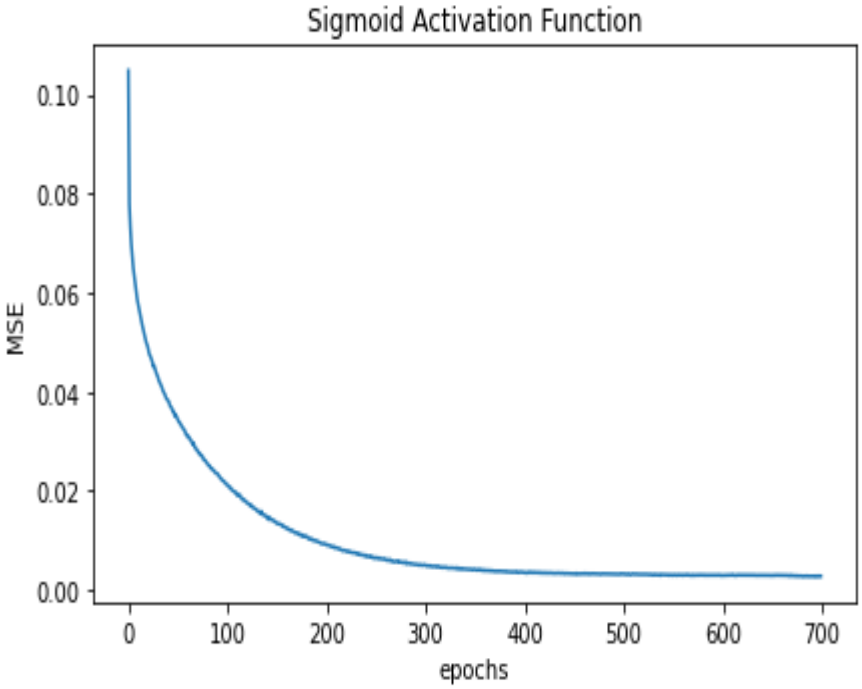
Inference

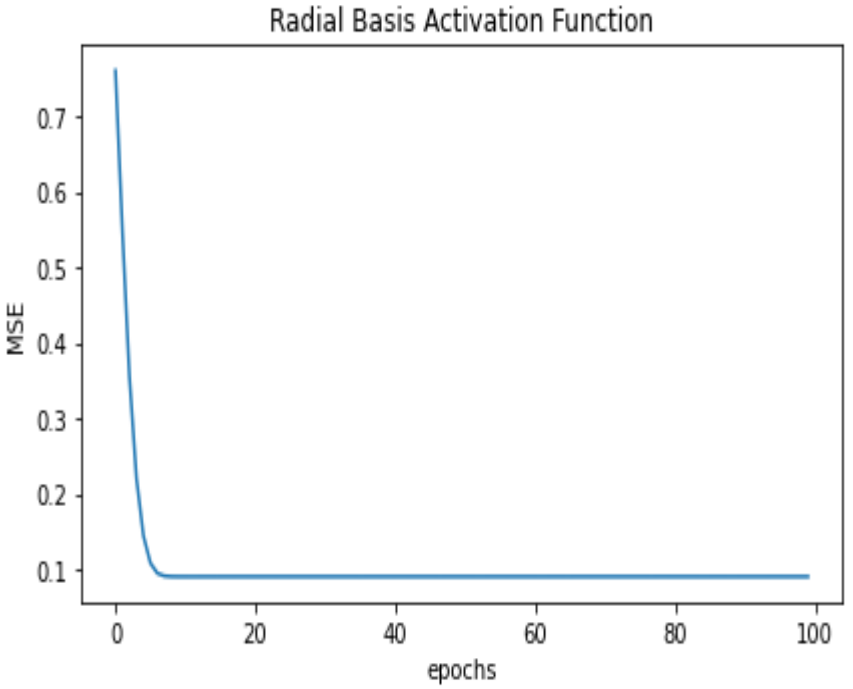
- RADIAL BASIS network.

The number of epochs given is 100 and the accuracy achieved is 6.5%

Results show that Radial basis function(NN) with single hidden layer is not suitable for Classification of Music genres

Plots – Epochs Vs MSE





References

- https://www.datacamp.com/community/tutorials/svm-classification-scikit-learn-python - **SVM**

(Theory) SVM,SVC

- SVM offers very high accuracy compared to other classifiers.
- The classifier separates data points using a hyperplane with the largest amount of margin and is also known as a discriminative classifier.
- SVM finds an optimal hyperplane which helps in classifying new data points.
- SVM constructs a hyperplane in multidimensional space to separate different classes and generates optimal hyperplane in an iterative manner, which is used to minimize an error.
- The core idea of SVM is to find a maximum marginal hyperplane(MMH) that best divides the dataset into classes.

(Theory) SVM,SVC

SVM Kernels

The SVM algorithm is implemented in practice using a kernel.

- **Linear Kernel** - Used as normal dot product any two given observations.
- **Radial Basis Function Kernel** - RBF can map an input space in infinite dimensional space.
- **Polynomial Kernel** - Distinguishes curved or nonlinear input space.
- gamma is a parameter, which ranges from 0 to 1. A higher value of gamma will perfectly fit the training dataset, which causes over-fitting. gamma = 0.1 is considered to be a good default value. The value of gamma needs to be manually specified in the learning algorithm.

Inference

- **SVC algorithm with polynomial kernel gives the highest accuracy score of 60.5%**
- **SVC algorithm with RBF kernel performs poorly for Music genre classification**

Accuracy Score

```
Accuracy of the Model (kernel='linear', gamma='0.1'): 56.5%
Accuracy of the Model (kernel='rbf'): 43.0%
Accuracy of the Model (gamma = 0.1, kernel='poly'): 60.5%
Accuracy of the Model (gamma=0.1): 11.0%
```

References

- https://www.analyticsvidhya.com/blog/2021/04/beginners-guide-to-decision-tree-classification-using-python/
- https://medium.com/swlh/decision-tree-classification-de64fc4d5aac

(Theory) Classification Tree

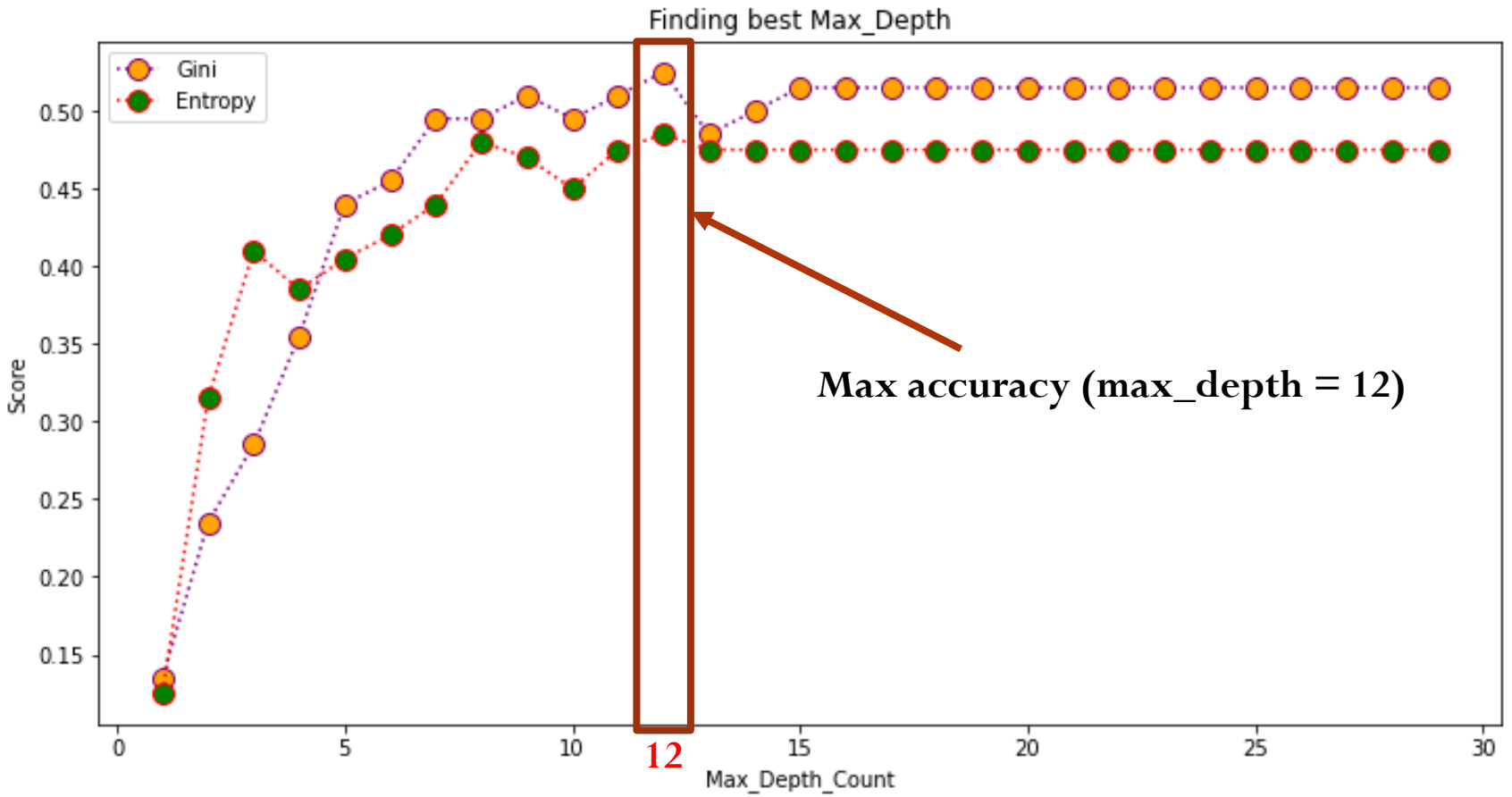
- Decision Tree is a Supervised learning technique that can be used for both classification and Regression problems, but it is preferred for solving Classification problems.
- It is a tree-structured classifier, where internal nodes represent the features of a dataset, branches represent the decision rules and each leaf node represents the outcome.
- It is a graphical representation for getting all the possible solutions to a problem/decision based on given conditions.

The two popular techniques for Attribute Selection Methods are:

- Information Gain (Entropy)
- Gini Index

Inference – DT Classification

Comparison of Gini Index and Entropy Criteria



Inference

From the above plot,

- **When the max_depth is 12, the accuracy of the Model (criterion as Gini index) : 52.5%**
- **When the max_depth is 12 ,the accuracy of the Model (criterion as Entropy) : 48.5%**

The Gini Index works better than Entropy for Music Genre Classification

References

Random Forest Algorithm

- https://www.askpython.com/python/examples/random-forest-regression
- https://machinelearningmastery.com/random-forest-ensemble-in-python/

(Theory) Random Forests

- Random forest is an ensemble of decision tree algorithms.
- It is an extension of bootstrap aggregation (bagging) of decision trees and can be used for classification and regression problems.
- In bagging, a number of decision trees are created where each tree is created from a different bootstrap sample of the training dataset.
- A bootstrap sample is a sample of the training dataset where a sample may appear more than once in the sample, referred to as sampling with replacement.
- Bagging is an effective ensemble algorithm as each decision tree is fit on a slightly different training dataset, and in turn, has a slightly different performance.

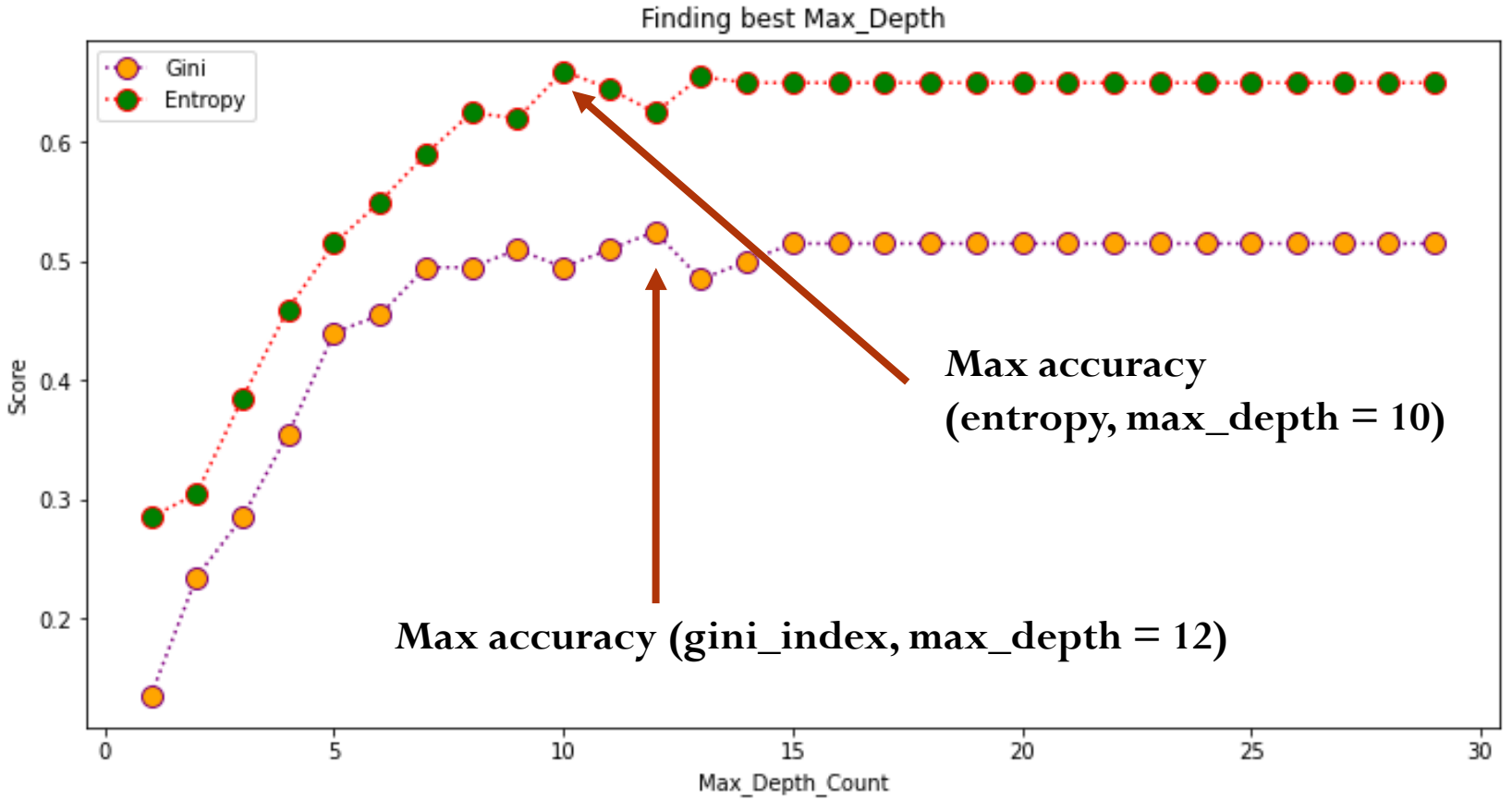
(Theory) Random Forests

Steps to perform the random forest regression

- Pick a random K data points from the training set.
- Build the decision tree associated to these K data points.
- Choose the number N tree of trees you want to build and repeat steps 1 and 2.
- For a new data point, make each one of your Ntree trees predict the value of Y for the data point in the question, and assign the new data point the average across all of the predicted Y values.

Inference

Comparison of Gini Index and Entropy Criteria



Inference

Comparing the models (Gini-index and entropy) of Decision tree and Random Forest algorithms

Algorithms	Gini _index		Entropy	
	Max accuracy	Max depth	Max accuracy	Max depth
Decision Tree	52.5 %	12	48.5 %	12
Random Forest	52.5 %	12	66 %	10

The maximum accuracy score for Music genre classification is 66 % (at max_depth = 10) and it is achieved using "Random Forest algorithm"